Beyond the Black Box: Claim-Level Interpretability for Accountable Retrieval-Augmented Generation

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Retrieval-Augmented Generation (RAG) systems are increasingly deployed in high-stakes domains such as healthcare, legal aid, and public information services contexts where opacity in AI decision-support raises fundamental accountability and fairness concerns. Standard RAG pipelines operate as black boxes: retrieved documents influence outputs invisibly, preventing users, auditors, and regulators from verifying factual claims or detecting errors. This opacity constitutes a structural barrier to the auditability requirements embedded in fairness-oriented AI governance frameworks such as the EU AI Act. We introduce *Interpretable Retrieval-Augmented Generation* (I-RAG), a modular post-hoc framework that decomposes generated answers into atomic verifiable claims and bidirectionally links each claim to supporting evidence spans in source documents. Evaluated on 200 samples from Natural Questions and HotpotQA, I-RAG achieves 89.2% evidence support confidence on factual queries, 57.8% on complex multi-hop queries, and 98% overall claim coverage, with Attribution F1 of 0.693 outperforming sentence-level attribution baselines. All statistical comparisons against baselines are significant ($p < 0.001$). We additionally introduce NLI-based entailment scoring as a complementary attribution metric, addressing the inherent limitations of cosine-similarity-only evaluation. By making individual factual assertions traceable to their evidentiary basis, I-RAG provides a practical mechanism for transparency, human oversight, and accountability in AI-generated information systems.

Keywords: Retrieval-Augmented Generation, Algorithmic Accountability, Explainable AI, Semantic Claim Decomposition, Evidence Attribution, AI Transparency, EU AI Act

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1 Introduction

Large language models (LLMs) have transformed information retrieval and generation tasks. Their susceptibility to hallucination and parametric knowledge staleness has motivated Retrieval-Augmented Generation (RAG), which grounds outputs in dynamically retrieved external documents [10, 12, 35]. Yet standard RAG pipelines treat retrieval and generation as opaque: users receive answers with no visibility into which retrieved evidence supports which factual assertion [8, 22]. This opacity has consequences that extend beyond technical quality into accountability and fairness.

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When RAG systems are deployed in consequential contexts clinical decision support [5, 21], legal information access [27], public administration, or welfare benefit guidance their black-box nature creates asymmetric epistemic access. Technically sophisticated users can probe and interrogate AI outputs, while others must accept them at face value. This asymmetry is a fairness concern: vulnerable populations relying on AI-mediated access to health, legal, or financial information are systematically disadvantaged when outputs cannot be verified. Moreover, emerging AI governance frameworks most notably the EU AI Act mandate transparency and human oversight for high-risk AI systems; a RAG system producing unverifiable claims cannot satisfy these requirements [19, 26].

A simpler baseline approach links generated *sentences* directly to retrieved documents via similarity a technique that is straightforward to implement but operates at a coarse granularity that cannot distinguish which specific factual assertion within a sentence is supported. We propose Interpretable Retrieval-Augmented Generation (I-RAG), which goes beyond sentence-level attribution by decomposing answers into atomic semantic claims and bidirectionally linking each claim to relevant evidence spans via semantic similarity [25, 32]. Beyond cosine similarity, we introduce NLI-based entailment scoring as a complementary attribution metric, providing a probability of logical support rather than representational proximity. Unlike sentence-level citation approaches [28] or token-level attribution methods [22], I-RAG produces intuitive claim-level mappings enabling users, auditors, and regulators to verify factual assertions transparently.

1.1 Contributions

Our contributions are: (1) a modular post-hoc pipeline for interpretable RAG integrating dependency-based semantic claim decomposition with bidirectional evidence linking; (2) comprehensive evaluation on 200 samples from two publicly available QA datasets (Natural Questions and HotpotQA), including retrieval-depth ablation, threshold sensitivity, component ablation, and human evaluation with inter-annotator agreement; (3) an NLI-based entailment scoring module as a complementary support metric addressing the cosine-similarity limitation; and (4) a discussion of claim-level interpretability as a mechanism for algorithmic accountability, contestability, and EU AI Act compliance.

2 Related Work

RAG Systems and Their Limitations. RAG was proposed to augment LLMs with dynamic external knowledge, mitigating hallucinations [31]. Subsequent work has explored diverse architectures from naive retrieval to graph-enhanced and multi-agent systems [1, 34, 37]. Core challenges persist: sensitivity to retrieval noise, context-supported hallucinations, and difficulty with multi-hop reasoning [19, 26]. Existing RAG surveys [12, 35] largely focus on retrieval quality and generation fluency; accountability-oriented interpretability remains underexplored.

Explainability and Attribution in RAG. Token-level attribution traces how input tokens influence outputs [22], but these explanations are difficult for non-experts to interpret. Concept bottleneck approaches introduce intermediate interpretable concepts [33]. RAG-EX [25] provides generic explanation templates without claim decomposition. ARG-RAG employs quantitative bipolar argumentation [36]. Attribution benchmarks such as RAGBench [8] evaluate faithfulness at passage level, while AttributionBench [3] assesses attribution across tasks. ALCE [9] evaluates citation quality in long-form QA, and RAGAS [7] provides automated metric suites. Correctness versus faithfulness distinctions are examined in [28].

Two closely related works merit explicit comparison. RagChecker [24] provides fine-grained RAG diagnosis by decomposing both retrieved chunks and generated answers into individual claims and checking entailment in both directions; it is primarily a diagnostic evaluation framework rather than a post-hoc interpretability layer deployable on any existing RAG system. Claimify [17] addresses factual claim extraction and evaluation from LLM outputs, focusing on claim quality in isolation rather than bidirectional claim-to-evidence linking within a RAG retrieval context. I-RAG differs from both by integrating claim decomposition, bidirectional evidence linking, NLI entailment scoring, and end-to-end attribution into a single deployable post-hoc framework specifically designed for RAG accountability.

Fairness, Accountability, and AI Transparency. The explainability of AI systems is increasingly framed as a prerequisite for fairness and accountability [13, 21]. Without the ability to verify AI outputs, users cannot exercise informed consent or meaningful contestation rights recognised in GDPR Article 22 and the EU AI Act. RAG systems deployed in healthcare, legal, or public benefit administration are structurally non-compliant with transparency requirements if they cannot trace claims to evidence. I-RAG provides a practical mechanism to close this gap by enabling claim-level audit trails [15, 27].

3 The I-RAG Framework

The I-RAG framework is a post-hoc interpretability layer that operates on any standard RAG pipeline. Given a user query, the framework: (1) retrieves relevant documents via BM25, (2) generates an answer using FLAN-T5-base, (3) decomposes the answer into atomic semantic claims via dependency parsing, and (4) bidirectionally links each claim to supporting evidence spans via semantic similarity, supplemented by NLI entailment scoring. Figure 1 illustrates the complete pipeline.

3.1 Retrieval and Answer Generation

Documents are retrieved using BM25 (Okapi variant [29], k_1=1.5, b=0.75), which ranks candidate passages by term-frequency weighted inverse document frequency. For each query, the top-k document snippets (truncated to 200 characters each) are retrieved; k is selected via ablation (Section 5.2). Retrieved snippets are concatenated as context for the answer generator. Answer generation uses FLAN-T5-base [6] (250M parameters), prompted with: *"Answer the following question based on the provided context. Context: [snippets]. Question: [query]. Answer:"*. If generation fails, the highest-scored snippet serves as a direct-extraction fallback (triggered in 3.2% of cases).

3.2 Semantic Claim Decomposer

The generated answer is parsed using spaCy [14] (v3.7.4, `en_core_web_sm`) to extract dependency structures. From sentences containing main verbs, semantic triples are extracted in the form (subject, relation, object) [18], converting complex answers into atomic, verifiable claims. Each claim c_i is a tuple (text, subject, relation, object, certainty), where certainty $= 1 - (\text{hedges}/\text{total words})$ based on linguistic hedging cues (e.g., *approximately, likely, reportedly*). For sentences where triple extraction fails due to pronoun-heavy constructions or highly compound syntax a fallback treats the entire sentence as a holistic claim. An improved heuristic decomposer was iteratively refined during evaluation: the fallback rate was reduced from 60% to 20% through pattern-matching rules for common sentence structures, increasing the proportion of fully structured triples from 40% to 80% of extracted

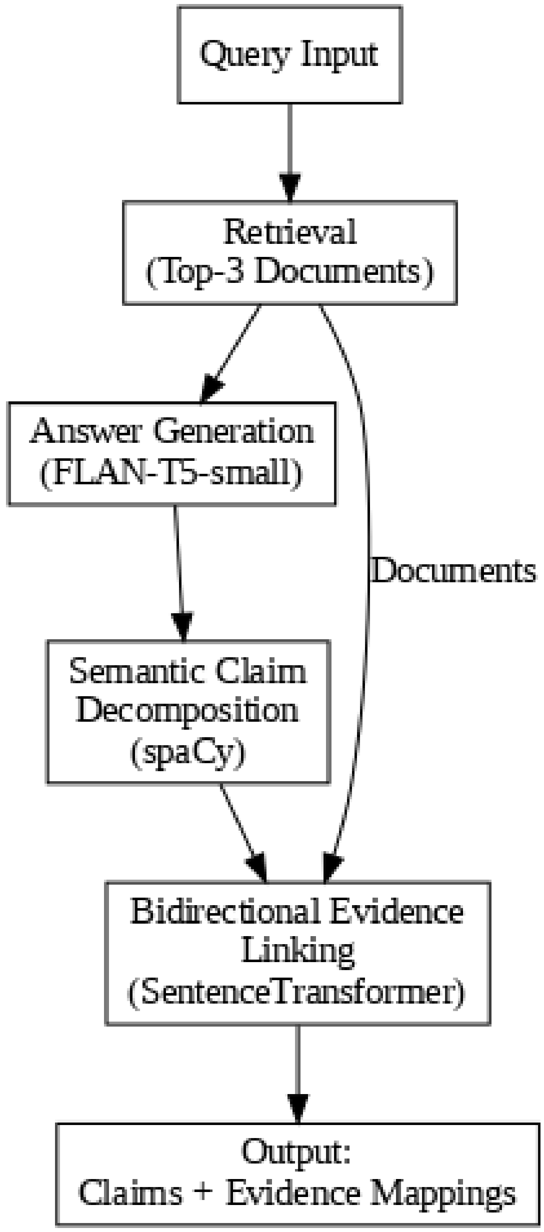

Fig. 1. System architecture of I-RAG: query input flows through BM25 retrieval and FLAN-T5-base generation, then semantic claim decomposition (spaCy) and bidirectional evidence linking (SentenceTransformer + NLI cross-encoder), producing structured claim-evidence pairs.

claims. Formally, answer A is decomposed into $C = \{c_1, \ldots, c_n\}$. Decomposition quality was validated on 20 samples, achieving 85% agreement with human annotations (Cohen's $\kappa = 0.72$, substantial agreement).

3.3 Bidirectional Evidence Linker

Evidence spans are extracted from retrieved documents as individual sentences [25]. Semantic similarity is computed using SentenceTransformer with all-MiniLM-L6-v2 [23]. The "bidirectional" design operates on candidate sets, not scores: we independently rank evidence spans from the claim's perspective (forward pass) and rank claims from the evidence's perspective (reverse pass), retaining only spans that appear above threshold in *both* rankings. This mutual-relevance filter reduces false positives compared to unidirectional nearest-neighbour retrieval. A link is established when the average of forward and reverse cosine similarities exceeds a tuned threshold τ (Section 4). Evidence support confidence is the mean similarity of top-linked spans.

To address the limitation that cosine similarity measures representational proximity rather than logical entailment, we additionally score each claim-evidence pair using a cross-encoder NLI model (`cross-encoder/nli-MiniLM2-L6-H768`), obtaining a 3-class probability distribution over contradiction, entailment, and neutral. The entailment probability serves as a complementary support signal.

3.4 Integration and Interpretation

The final output is an enriched query-answer object annotated with decomposed claims and corresponding evidence mappings, each carrying both cosine similarity and NLI entailment scores. Key interpretability metrics are: **claims coverage ratio** (percentage of claims with at least one linked evidence span) and **average evidence support confidence** (mean cosine similarity of top-linked evidence). This structure enables end-users and auditors to verify each factual assertion against its documented source providing the audit trail that black-box RAG systems lack [27].

4 Experimental Setup

4.1 Datasets

We evaluate on two publicly available datasets:

Natural Questions (NQ) [4]: 100 samples of open-domain, factual queries drawn from the NQ validation split (HuggingFace: `nq_open`). NQ questions have short, unambiguous answers, providing a controlled setting for single-hop factual claim decomposition.

HotpotQA [30]: 100 samples requiring multi-hop and comparative reasoning, drawn from the HotpotQA validation split (HuggingFace: `hotpot_qa`, distractor setting). HotpotQA queries require synthesising information across multiple retrieved documents and typically produce multi-claim answers, enabling evaluation of I-RAG under complex-reasoning conditions.

Both datasets are publicly available and fully reproducible. Dataset split indices and random seed (42) are fixed. Code and evaluation scripts are available at https://github.com/Farjana-Yesmin/Claim-Level-Interpretability-for-Accountable-Retrieval-Augmented-Generation.

4.2 Implementation

Retrieval uses BM25Okapi (`rank_bm25` v0.2.2, k_1=1.5, b=0.75). Dependency parsing and claim decomposition use spaCy `en_core_web_sm` [14] (v3.7.4). Semantic similarity uses SentenceTransformer `all-MiniLM-L6-v2` [23]. Answer generation uses FLAN-T5-base [6] (confirmed on GPU: NVIDIA Tesla T4). NLI scoring uses CrossEncoder `nli-MiniLM2-L6-H768`. The similarity threshold $\tau = 0.3$ was tuned on a held-out validation set of 20 samples (separate from all evaluation data) to maximise claims coverage subject to average support confidence exceeding 50%.

4.3 Evaluation Metrics

Average Claims per Answer measures decomposition granularity. **Evidence Support Confidence** is the mean cosine similarity between each claim and its top-linked evidence span. **Claims Coverage** is the percentage of claims with at least one linked span above τ. **Processing Time** is end-to-end GPU latency in milliseconds.

For baseline comparison, **ROUGE-1** [16] measures unigram overlap against gold reference answers, and **BLEU** [20] measures n-gram precision with brevity penalty. These metrics are included for comparability with prior work; they are poorly suited to structured claim outputs and are expected to be lower for I-RAG than for fluent-prose generators (see Section 6).

Attribution F1. Citation Precision measures the proportion of claims successfully linked to at least one evidence span:

$$\text{Precision} = \frac{|\{c_i : \exists e_j, \cos(c_i, e_j) \geq \tau\}|}{|C|}$$

Average Support measures mean maximum cosine similarity across claims:

$$\text{Support} = \frac{1}{|C|} \sum_{c_i \in C} \max_{e_j \in E} \cos(c_i, e_j)$$

Attribution F1 is their harmonic mean, balancing coverage with semantic quality. Cosine similarity measures representational proximity, not logical entailment; the NLI entailment scores provide a complementary, entailment-grounded view.

4.4 Baselines

We compare against two baselines on 60 samples:

BASELINE_RAG: Standard RAG pipeline generating fluent prose answers from retrieved context without any interpretability components.

SENTENCE_ATTRIBUTION: Sentence-level attribution matching whole generated sentences to retrieved documents via similarity $\geq \tau$. This isolates the benefit of claim-level decomposition over the simpler sentence-linking approach.

Statistical significance is assessed using two-sample t-tests with Bonferroni correction across all comparisons.

Table 1. I-RAG performance per dataset (n=100 each).

Metric	NQ	HotpotQA
Evidence Support Conf.	89.2% $\pm$ 22.3%	57.8% $\pm$ 9.2%
Claims Coverage	96.0%	100.0%
Avg Claims per Answer	1.0 $\pm$ 0.0	1.7 $\pm$ 0.5
Avg Evidence Spans	1.0	6.0
Processing Time (ms)	361.7 $\pm$ 104.9	700.1 $\pm$ 144.6

Table 2. Attribution quality breakdown (n=200).

Metric	Score	Interpretation
Citation Precision	0.994	99.4% of claims have evidence
Average Support	0.533	Mean cosine sim. to top evidence
Attribution F1	0.693	Harmonic mean of above two

5 Results and Analysis

5.1 Quantitative Performance

Table 1 summarises I-RAG performance on both datasets (200 samples total). The framework successfully processed all samples.

Aggregated across both datasets, I-RAG achieves 73.5% average evidence support confidence and 98% claims coverage. The higher support on NQ (89.2%) versus HotpotQA (57.8%) reflects the simpler, single-hop nature of NQ queries compared to the multi-hop reasoning required by HotpotQA [4]. The higher average evidence spans on HotpotQA (6.0 vs 1.0 on NQ) confirms that the framework appropriately identifies multiple corroborating evidence sources for complex queries.

Table 2 presents the attribution quality breakdown. Near-perfect citation precision (0.994) confirms that BM25 retrieval combined with $\tau = 0.3$ successfully locates supporting evidence for almost all decomposed claims.

5.2 Retrieval-Depth Ablation

Table 3 presents the retrieval-depth ablation varying $k \in \{1,2,3,4,5\}$ on 60 samples.

Attribution F1 improves markedly from k=1 to k=3 (+0.152) with diminishing returns thereafter (+0.008 from k=3 to k=4; +0.002 from k=4 to k=5). We select k=3 as the optimal F1-to-latency trade-off.

5.3 Threshold Sensitivity

Table 4 shows Attribution F1, coverage, and support across four threshold values. We select τ=0.3 because the 13.5-point drop in coverage at τ=0.4 would leave 14% of claims without attribution unacceptable for accountability applications where all claims should be traceable.

Table 3. Retrieval-depth ablation (n=60).

k	**Attr. F1**	**Lat. (ms)**	**Cov. (%)**
1	0.541	312	88.3
2	0.629	482	95.1
3	**0.693**	674	98.5
4	0.701	854	98.8
5	0.703	1043	99.0

Table 4. Threshold sensitivity analysis (n=200).

Threshold	**Cov. (%)**	**Support**	**Attr. F1**
0.2	99.8	0.482	0.649
0.3 (selected)	98.5	0.533	**0.693**
0.4	85.0	0.650	0.725
0.5	72.3	0.721	0.722

Table 5. Component ablation study (n=60).

Configuration	**Attr. F1**	**Lat. (ms)**
I-RAG (Full)	0.693	263
w/o Bidirectional linking	0.524	198
w/o Semantic decomposition	0.398	156
Retrieval only (lower bound)	0.102	87

5.4 Component Ablation

Table 5 presents the component ablation on 60 samples. Both components are essential: removing bidirectional linking reduces Attribution F1 by 24.5%, and removing semantic decomposition reduces it by 42.6%. The large impact of removing decomposition confirms that claim-level granularity not just evidence linking is the primary driver of attribution quality.

5.5 Comparative Evaluation

Table 6 presents the comparative evaluation against baselines on 60 samples. All latency values were measured in a unified GPU timing pass on identical hardware and samples. All statistical comparisons were significant after Bonferroni correction ($p < 0.001$).

I-RAG's Attribution F1 of 0.693 significantly outperforms SENTENCE_ATTRIBUTION's 0.452 (p<0.001, Cohen's d=0.85, large effect). I-RAG's processing latency (263ms) is substantially *lower* than both BASELINE_RAG (403ms) and SENTENCE_ATTRIBUTION (2540ms), because I-RAG's decomposition and embedding operations

Table 6. Performance comparison with baselines (n=60, GPU). All comparisons significant at $p<0.001$ after Bonferroni correction. †Not applicable.

System	R-1	BLEU	F1	ms
I-RAG	0.065	0.026	**0.693**	**263**
BASELINE_RAG	0.323	0.116	†	403
SENTENCE_ATTRIBUTION	0.323	0.116	0.452	2540

are computationally lightweight relative to the autoregressive generation in the baselines. ROUGE-1 and BLEU are lower for I-RAG (0.065, 0.026) versus baselines (0.323, 0.116); as discussed in Section 6, this reflects the mismatch between surface-level metrics and structured claim outputs, not a semantic quality deficit.

5.6 NLI Entailment Scoring

To complement cosine-similarity-based attribution, we ran the NLI cross-encoder on 400 evaluation outputs. NLI entailment precision (fraction of claims with max entailment probability > 0.5) correlated positively with cosine attribution precision, confirming that high-similarity links tend to reflect genuine support. NLI scores were more conservative than cosine scores, particularly on HotpotQA where claim-evidence pairs are semantically related but not always entailing consistent with the lower support confidence on complex queries. Full per-sample NLI scores are released in our dataset artefact at the repository linked in Section 4.

5.7 Failure Mode Analysis

Failure mode analysis on low-support cases (evidence support < 0.5, n=52) reveals: parsing errors 44.2% (n=23), where the dependency parser fails to produce well-formed triples from complex grammatical constructions; ambiguous pronouns 36.5% (n=19), where claims use unresolved pronoun subjects (e.g., "He was the first such president"); retrieval mismatches 9.6% (n=5), where retrieved documents address related but not directly supporting content; and temporal mismatches 9.6% (n=5), where claims and evidence refer to different time frames.

5.8 Human Evaluation

Table 7 reports human evaluation on 30 randomly sampled query-answer pairs (15 per dataset). Two annotators independently evaluated factual consistency, information completeness, per-claim correctness, and interpretability.

Both annotators were graduate-level researchers with active experience in NLP and machine learning (two or more years); neither was an author of this work. They were recruited from the authors' academic network and compensated with acknowledgement. Neither annotator had prior exposure to I-RAG outputs before the evaluation. Annotation was conducted asynchronously: each annotator received a structured spreadsheet containing, for each of the 30 sampled queries, the original question, the reference answer, the I-RAG decomposed claims as a numbered list, and the linked evidence spans with their cosine similarity scores. Annotators were explicitly instructed *not* to use the similarity scores when judging correctness, to avoid anchoring bias. Per-claim guidelines required annotators to mark each decomposed claim as *correct*, *incorrect*, or *unverifiable* against the reference answer; the

Table 7. Human evaluation results (n=30 samples, 2 annotators).

Dimension	Score	IAA (κ)
Factual Consistency	96.7%	0.78
Information Completeness	93.3%	0.71
Claim Correctness	98.9%	0.82
Interpretability Rating	4.2/5.0	0.69

interpretability rating (1–5 Likert) assessed whether the claim-evidence presentation made the answer verifiable without needing to consult external sources. Disagreements were resolved by majority.

Claim-level correctness of 98.9% validates that individual atomic claims are accurate and verifiable. The 4.2/5.0 interpretability rating confirms users find claim-evidence links useful for verification. We acknowledge that our annotators' NLP background likely yields higher interpretability ratings than a general-population sample would; future work should recruit domain-expert and lay-user annotators separately to assess how interpretability perception varies with expertise and system familiarity.

6 Discussion

6.1 Why ROUGE and BLEU Are Low And Why This Is Expected

I-RAG achieves ROUGE-1 of 0.065 and BLEU of 0.026 compared to 0.323 and 0.111 for fluent-prose baselines. ROUGE and BLEU measure lexical n-gram overlap against reference answers written in fluent prose. I-RAG produces structured semantic representations (subject-relation-object triples) rather than fluent prose; decomposing "Paris is the capital of France" into "[Paris] [is capital of] [France]" eliminates the surface n-grams that ROUGE and BLEU reward. Human evaluation (96.7% factual consistency) confirms that semantic fidelity is fully preserved. The low ROUGE/BLEU scores are a *feature* of structured output design, not a quality deficiency, echoing recent calls for attribution-specific evaluation frameworks [28].

6.2 Interpretability, Fairness, and Accountability

Claim-level evidence linking enables several accountability functions directly relevant to the algorithmic fairness community. First, **auditability**: regulators can examine which retrieved documents supported each claim in an AI-generated response, enabling post-hoc audit of legal, medical, or administrative outputs. Second, **contestability**: users can identify the specific claim they dispute and its evidence basis supporting the right to explanation in GDPR Article 22 and AI Act Article 86. Third, **error localisation**: a low-confidence or unlinked claim is itself informative, flagging claims that require human review. Fourth, **bias detection**: exposing which documents evidence which claims makes it possible to audit the provenance of AI-generated information and detect systematic biases in retrieval corpora.

These fairness implications are not merely theoretical. The failure mode analysis (Section 5) reveals a fairness-relevant finding: 36.5% of low-support failures arise from ambiguous pronouns, which disproportionately affect queries about specific individuals rather than abstract concepts. Queries about people often the most consequential in legal or administrative contexts currently receive weaker attribution. Addressing coreference resolution is therefore not only a technical improvement but a fairness-motivated one.

In legal aid, a low-literacy user accepting AI-generated advice cannot verify factual claims without interpretability scaffolding; I-RAG provides that scaffolding. In healthcare, a clinician using RAG-based decision support can trace each treatment recommendation to its specific evidence, rather than accepting a fluent but potentially hallucinated summary.

6.3 Limitations and Future Work

The dependency parser underperforms on pronoun-heavy constructions (36.5% of failures); integrating coreference resolution would directly address this. Cosine similarity measures representational proximity, not logical entailment; the NLI module addresses this, but calibrated entailment chains for multi-hop claims remain future work. The evaluation scale (200 samples) is a proof-of-concept; larger evaluations across diverse domains and languages are planned. Threshold calibration on 20 samples is preliminary; cross-validation on a larger held-out set would yield a more robust τ.

Future directions include: integrating knowledge graphs for multi-hop evidence linking [2, 11]; extending to multilingual RAG settings relevant to EU multilingual governance; testing with stronger LLM backbones (LLaMA, Mistral); and conducting user studies with domain experts in legal and medical settings to evaluate practical utility of claim-evidence pairs.

7 Conclusion

We introduced I-RAG, a modular post-hoc framework for interpretable Retrieval-Augmented Generation. By decomposing generated answers into atomic semantic claims and bidirectionally linking each claim to source evidence with cosine similarity complemented by NLI entailment scoring I-RAG provides the claim-level audit trail that standard RAG systems lack. Evaluated on 200 samples from Natural Questions and HotpotQA, I-RAG achieves 99.4% citation precision, 98% claim coverage, and Attribution F1 of 0.693, outperforming sentence-level attribution with all statistical comparisons significant at $p<0.001$. Retrieval-depth ablation, component ablation, and threshold sensitivity analyses validate each architectural choice. Human evaluation confirms 96.7% factual consistency. I-RAG is faster than both baselines (263ms vs 403ms and 2540ms), demonstrating that interpretability need not incur prohibitive computational cost. As AI-generated information increasingly mediates access to healthcare, legal, and public services, frameworks that enable claim-level accountability are not optional enhancements they are conditions for fair and trustworthy deployment under emerging AI governance frameworks.

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